

THE 3-GATE MORPHOLOGICAL REACHABILITY & SINGULARITY VALIDATOR

Deterministic screening of untrusted VLM stochastic proposals before granting physical motor execution authority

UNTRUSTED INPUT PROPOSAL (VLM /LLM SYSTEM 2)

Candidate Target Pose: $T_{target} = [R \mid p] \in SE(3)$ | Payload: $m = 0.50$ kg | Origin: Linux MPU (Untrusted)



GATE 1: WORKSPACE GEOMETRY (W_{reach})

$$r_{min} \leq ||p|| \leq r_{max}$$

$$0.15 \text{ m} \leq r \leq 0.85 \text{ m}$$

$$z_{target} \geq z_{table} + 20 \text{ mm}$$

- **Workspace Sphere:** Rejects out-of-reach targets.
- **Table Plane Hull:** Prevents surface collision.
- **Joint Range Check:** Validates $q_{min} \leq IK(T) \leq q_{max}$.

✓ PASSED: WITHIN DEXTEROUS WORKSPACE

GATE 2: SINGULARITY & MANIPULABILITY

$$\kappa(J) = \sigma_{max} / \sigma_{min} \leq 50.0$$

$$w(q) = \sqrt{(\det(J J^T))} \geq 0.05$$

$$\det(J) \neq 0 \text{ (No Rank Loss)}$$

- **Singularity Avoidance:** Prevents velocity spikes ($\dot{q} = J^{-1} \dot{x} \rightarrow \infty$) near boundary configurations.
- **Yoshikawa Ellipsoid:** Ensures non-zero dexterity.

✓ PASSED: CONDITION NUMBER $\kappa(J) = 14.2$

GATE 3: STATIC TORQUE & PAYLOAD

$$\tau_{static} = g(q) + J^T F_{ext}$$

$$|\tau_i| \leq 0.80 \cdot \tau_{continuous}$$

Thermal Margin: 20% Headroom

- **Gravity Holding:** Computes required motor torque.
- **Thermal Protection:** Prevents winding $I^2 R$ Joule overheating during sustained grasp holds.

✓ PASSED: PEAK TORQUE 62% OF RATED LIMIT

GATEWAY AUDIT COMPLETE: ALL 3 MORPHOLOGICAL GATES SATISFIED

- **Permission Granted:** Proposal certified as kinematically admissible, singularity-safe, and within motor thermal limits.
- **Handoff to System 1.5:** Target pose T_{target} is passed to the real-time trajectory planner to synthesize C^2 -continuous action chunks.
- **Defect Reporting:** If any gate had failed, an explicit error code (e.g. ERR_SINGULARITY_KAPPA) would be returned to System 2 for re-prompting.